

Vol 12 Chapter 1 — The Periodic Table from Substrate

Keeper (author pass — deep math/physics revision)

2026-05-24 Sunday

Chapter 1 — The Periodic Table from Substrate

Level 1 — one sentence

The periodic table — Mendeleev's organizing structure for all chemistry — emerges in BST as the substrate's natural multi-electron atomic K-type shell structure, with orbital degeneracy sequence $(2\ell + 1) = 1, 3, 5, 7$ matching BST primary integers $\{1, N_c, n_C, g\}$ (Vol 5 Ch 3), Madelung filling order, and $N_{\max} = 137$ as the substrate-derived stability ceiling for elements.

Level 2 — graduate-physicist precision

1.1 The periodic table

Mendeleev 1869 / Meyer 1870: arranged elements by atomic weight; periodic recurrence of chemical properties. Refined to atomic number after Moseley 1913.

Shell-cumulative capacities: $2, 8, 18, 32 = 2 \cdot 1, 2 \cdot (1 + 3), 2 \cdot (1 + 3 + 5), 2 \cdot (1 + 3 + 5 + 7)$. (Each capacity recurs across two periodic-table rows: row 1 holds 2, rows 2-3 hold 8 each, rows 4-5 hold 18 each, rows 6-7 hold 32 each — per Madelung filling.)

Madelung filling order: 1s, 2s, 2p, 3s, 3p, 4s, 3d, 4p, 5s, 4d, 5p, 6s, 4f, 5d, 6p, 7s, 5f, 6d, 7p, ...

1.2 Substrate K-type derivation

Vol 5 Ch 3 (substrate angular momentum + Pin(2) spin): orbital degeneracy $(2\ell + 1) = 1, 3, 5, 7$ matches BST primary sequence $\{1, N_c, n_C, g\}$. Spin doubling $\times (2\ell + 1)$ gives shell capacities 2, 6, 10, 14 for $\ell = 0, 1, 2, 3$ (s, p, d, f).

Shell-cumulative capacities $2, 8, 18, 32 =$ cumulative shell sums (Madelung). The substrate's $SO(5)$ K-type degeneracy structure $\times$ Pin(2) spin doubling = periodic-table structure.

1.3 $N_{\max} = 137 =$ substrate stability ceiling

$$N_{\max} = N_c^3 \cdot n_C + \text{rank} = 27 \cdot 5 + 2 = 137.$$

Independent confirmation via Dirac $Z \cdot \alpha = 1$ limit (Grace INV-5123 literature scan Sunday 2026-05-24): the BST primary identity $N_{\max} = 137 = 1/\alpha$ has solid theoretical anchor in Dirac's Feynman-limit $Z \cdot \alpha = 1$ for atomic stability — when the strongest 1s-electron orbital reaches relativistic $Z \cdot \alpha \approx 1$, hydrogenic perturbation theory breaks down. Standard textbook result: $Z = 1/\alpha \approx 137$ is the stability ceiling for hydrogen-like atoms. BST's $N_{\max} = 137$ identity is not numerology — it is the three-way intersection BST primary integer $\leftrightarrow$ Dirac- $Z \leftrightarrow 1/\alpha$.

For elements: $Z \leq N_{\max}$ predicted stable. $Z > 137$: relativistic + substrate K-type structure both fail.

Current periodic table: $Z = 1$ (hydrogen) to $Z = 118$ (oganesson, named 2016). $Z = 119, 120, \dots$ being sought. BST predicts stability degradation as $Z \rightarrow 137$ and instability for $Z > 137$ — consistent with experimental synthesis status (no element synthesized past 118 yet; 137 not approached).

Task #111 (BST superheavy nuclei island stability) supports this.

1.4 Chemistry from substrate

All chemistry follows from substrate atomic K-type structure: - Bonding (Ch 2): substrate K-type overlap - Molecular orbitals (Ch 3): substrate K-type linear combinations - Reactions (Ch 4): substrate K-type rearrangements - Spectroscopy (Ch 6): substrate K-type transitions

1.5 K-audit anchors

- Vol 5 Ch 3: SO(5) K-types, orbital sequence
- Vol 0 Ch 5: $N_{\max} = 137$ derivation
- Task #111 completed: superheavy nuclei

Level 3 — 5th-grader accessibility

Periodic table: 118 elements arranged in 7 rows. Shell-cumulative capacities 2, 8, 18, 32 (each capacity recurs across two rows). These come from $(2\ell + 1)$ sequence: 1, 3, 5, 7 — where the three non-trivial degeneracies $\{3, 5, 7\} = \{N_c, n_c, g\}$ are BST primaries. Spin doubling $\times$ orbital degeneracy $\rightarrow$ shell capacities. $N_{\max} = 137$: substrate stability ceiling for elements. Beyond $Z = 137$: substrate can't hold the atom together.

What comes next

Chapter 2 develops chemical bonding from substrate.

Where to look this up

- Standard chemistry texts
- BST: Vol 5 Ch 3; Vol 0 Ch 5